

Assembly and wire connections

HOME-ETCH PROTOTYPE | Front-view coordinates measured from the top-left corner

Two copper faces; no plated holes or solder mask are assumed.

Fit surface-mount electronics. RGB LEDs, piezo and the display header are through-hole.

Install 30 short via wires and solder BOTH faces. Do not rely on an unplated hole.

Fit the five insulated links below. They land on existing component solder joints.

Link lines in the drawings identify endpoints; route the wires clear of LED/display faces.

Fit vias before the piezo; leave clearance beneath its body for solder joints.

Required insulated wire links

WL1	U2.4 to R5.1 Front direct span 26.2 mm
WL2	U2.12 to U1.12 Front direct span 31.0 mm
WL3	U2.14 to U1.9 Front direct span 24.7 mm
WL4	U2.15 to R8.1 Front direct span 33.3 mm
WL5	J2.2 to J3.3 Rear direct span 22.3 mm

These are real assembly wires, not copper traces or optional links.

KiCad models their fitted connectivity as board-only jumper footprints.

The connection checks pass only for the assembly with all five links fitted.

Drilling and mechanical fit

LEDs, piezo, display header and vias: 1.0 mm holes.

J3 terminal: 1.1 mm holes. USB-A shell tabs: 2.3114 mm holes; fit the actual tabs.

RGB lead holes: 2.54 mm pitch, 2.0 mm copper pads; form leads if necessary.

Piezo lead spacing remains the selected footprint's 10.0 mm; verify the actual part.

Display: 51 x 28 mm envelope, five-wire connection, no guessed mounting holes.

Support and insulate the display above solder joints. Check standoff height first.

J3 body overhangs the right board edge by about 1.4 mm, plus wire clearance.

Power and first checks

J3.1 = regulated 5 V input; J3.4 = GND. Power XIAO separately through USB-C.

USB-A output is custom 3.3 V (100-200 mA normal, 500 mA transient).

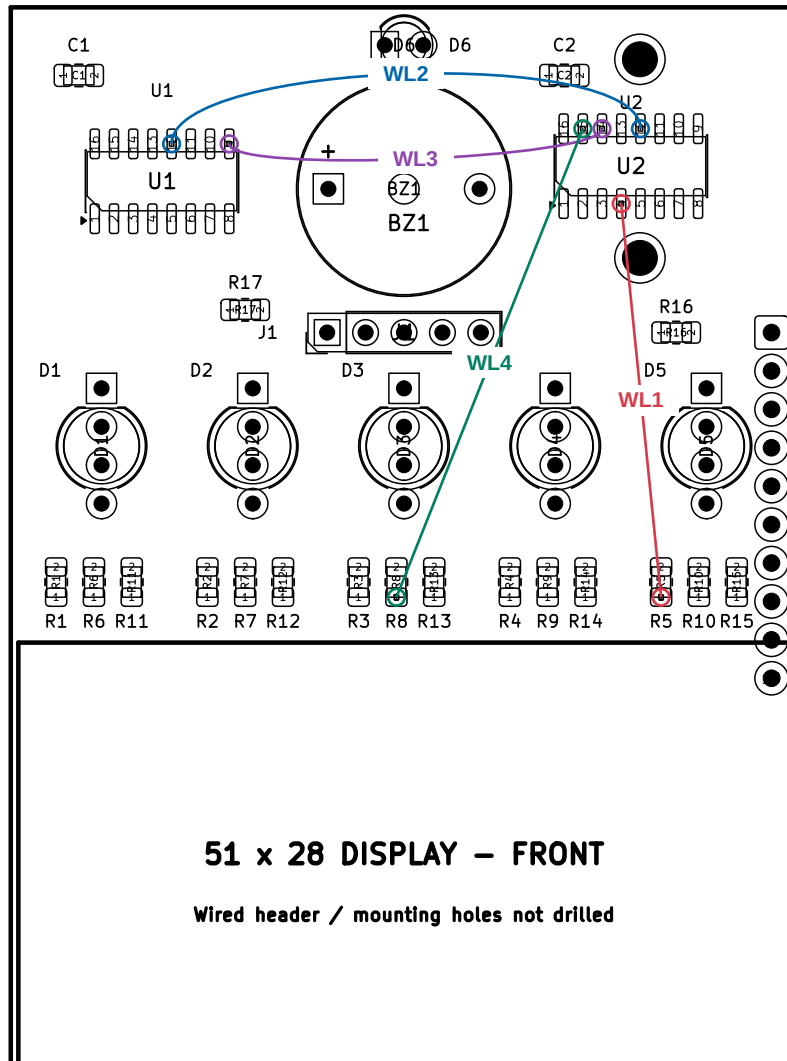
Display header left-to-right: VI2C (3.3 V), 5 V, GND, SDA, SCL.

Inspect both faces for bridges and continuity before applying current-limited power.

D+/D- wiring is not a qualified controlled-impedance USB link.

Front assembly

2:1 inspection drawing - NOT toner artwork. Connector openings face right from the front.



All component references are shown for assembly; there is no assumed silkscreen.

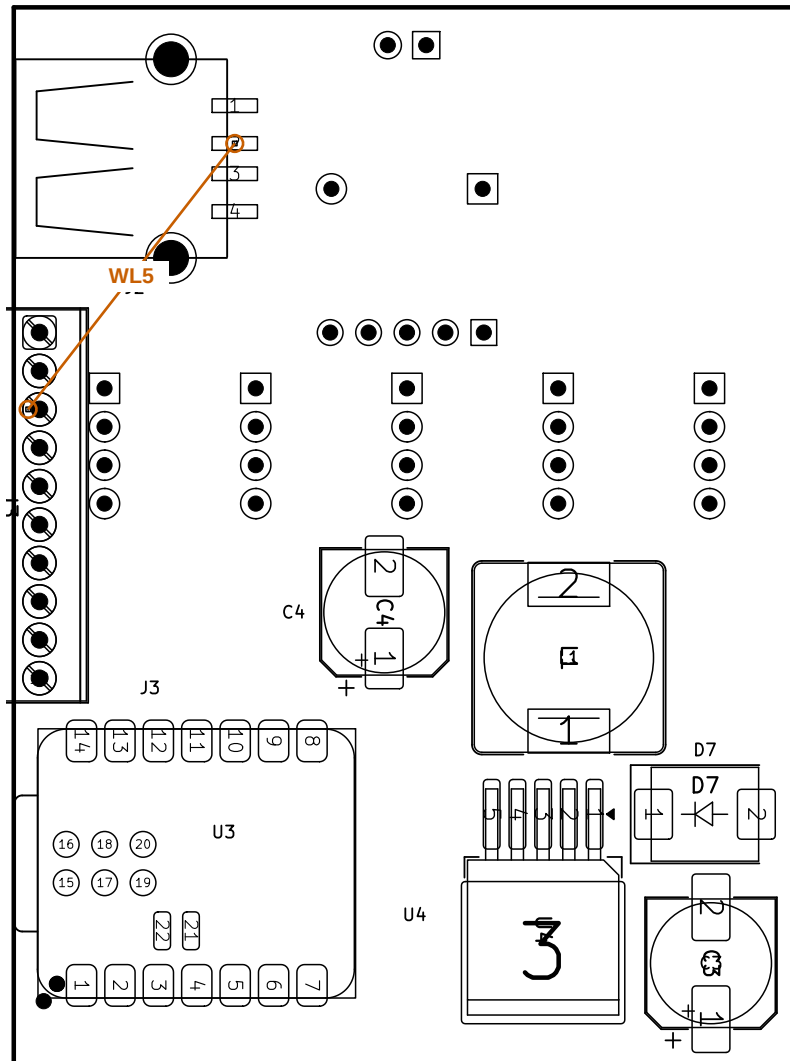
Front LED body centres: x = 6, 16, 26, 36, 46 mm; y = 29 mm.

RGB physical lead order from top to bottom: A, R, G, B.

Display envelope starts at x = 0.5 mm, y = 42 mm; mounting holes are omitted.

Rear assembly

2:1 inspection drawing - NOT toner artwork. Connector openings face right from the front.



All component references are shown for assembly; there is no assumed silkscreen.

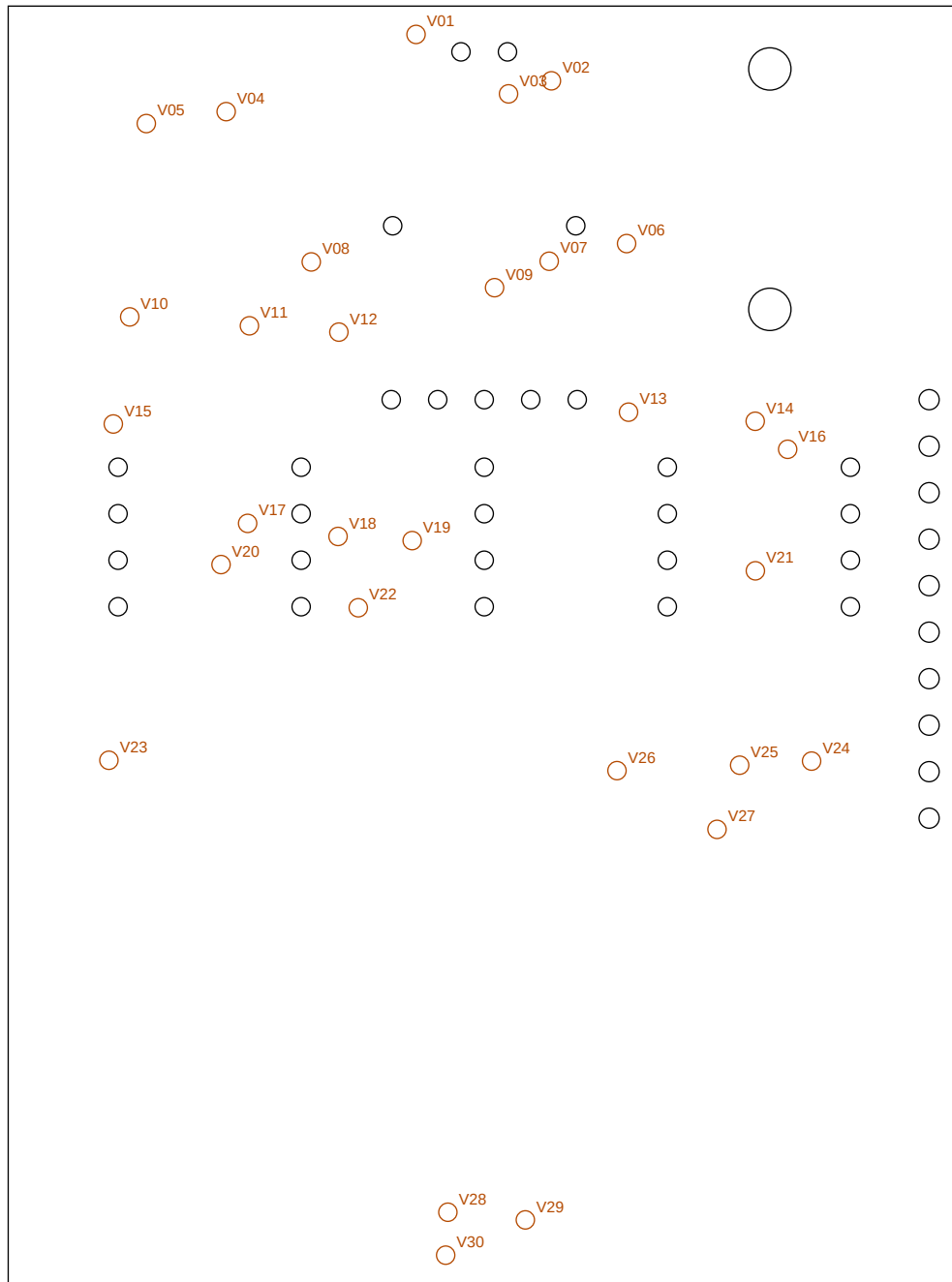
Front LED body centres: x = 6, 16, 26, 36, 46 mm; y = 29 mm.

RGB physical lead order from top to bottom: A, R, G, B.

Fit the XIAO and connectors only after the nearby via wires are soldered and inspected.

Drill and via map

Front view | 2.5:1 inspection drawing - use the copper print sheets for transfer



Black circles: component holes. Orange numbered circles: 1 mm wire-filled vias.
Vias have 2 mm copper pads. Solder a wire on both faces at every orange location.
The next page gives exact via coordinates. No mounting holes are included.

Via-wire checklist

All positions in mm from the front top-left corner. All via drills are 1.00 mm.

Via	X	Y	Done	Via	X	Y	Done
V01	22.276	1.548	☐	V16	42.573	24.210	☐
V02	29.669	4.082	☐	V17	13.077	28.257	☐
V03	27.328	4.795	☐	V18	18.016	28.971	☐
V04	11.905	5.766	☐	V19	22.066	29.199	☐
V05	7.543	6.417	☐	V20	11.625	30.507	☐
V06	33.776	12.973	☐	V21	40.810	30.849	☐
V07	29.548	13.937	☐	V22	19.121	32.867	☐
V08	16.553	13.975	☐	V23	5.500	41.198	☐
V09	26.565	15.382	☐	V24	43.880	41.243	☐
V10	6.637	16.979	☐	V25	39.952	41.468	☐
V11	13.175	17.465	☐	V26	33.251	41.764	☐
V12	18.061	17.809	☐	V27	38.715	44.973	☐
V13	33.882	22.182	☐	V28	24.014	65.887	☐
V14	40.801	22.679	☐	V29	28.246	66.306	☐
V15	5.741	22.829	☐	V30	23.893	68.233	☐

- Trim the wire close after soldering, without cutting into the copper pad.
- Check continuity from the front trace to the rear trace; visual solder alone is not proof.
- Also solder through-hole component leads on each face that carries a connection.
- Do not substitute a bare hole for a via or omit a required insulated jumper.